

Step 10

⇒ Flight Path Angle Calc. for Rocket Launch

$$\gamma = \arcsin\left(\frac{\vec{v} \cdot \vec{r}}{\|\vec{v}\| \|\vec{r}\|}\right) = \frac{\pi}{2} - \arccos\left(\frac{\vec{v} \cdot \vec{r}}{\|\vec{v}\| \|\vec{r}\|}\right)$$

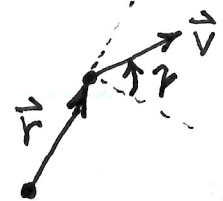
where

$\vec{v}$ is either:

$\vec{v}_{inertial}$, used above 80 km

or

$\vec{v}_{rel} = \vec{v}_{inertial} - (\vec{\omega}_{\oplus} \times \vec{r})$, used below 40 km



- Can apply fxn to blend use of $\vec{v}_{inertial}$ & $\vec{v}_{rel}$ betw 40 km & 80 km.

- To implement continuous blending w/o sudden steering "jerk", use smooth step fxn based on altitude (h):

use Cubic hermite spline to ensure smooth derivatives (velocities & accelerations) at transition edges

$$\ast \vec{v}_{steer}(h) = (1 - w(h)) \vec{v}_{rel} + w(h) \vec{v}_{inertial}$$

where

$w(h)$ = weight factor in range $[0, 1]$

Define transition window using lower alt. limit (h_{min}) & upper alt. limit (h_{max})

Standard Earth parameters:

$$h_{min} = 40 \text{ km}$$

$$h_{max} = 80 \text{ km}$$

Normalize alt. into $[0, 1]$ range:

$$x = \frac{h - h_{min}}{h_{max} - h_{min}}$$

(clamp x to ensure its within $[0, 1]$ range:

$$x = \max(0, \min(1, x))$$



(Formulation of $w(h)$ in $\vec{v}_{steer}$ eqn cont.)
Apply the cubic hermite spline to get $w(h)$:

$$w(h) = 3x^2 - 2x^3$$

† This ensures $\frac{dw}{dh} = 0$ at both h_{min} & h_{max} .

> Note: (E.g. q , F_D , M)

Always compute aerodynamic physics ^strictly using the relative wind frame.

Blending fxn used to compute $\vec{v}_{steer}$, used in computing γ (& the η required to achieve γ at burnout)
